

Grant Kaufmann

 grant-kaufmann-harvard  grantkaufmann@college.harvard.edu  <https://grantkaufmann88.github.io/portfolio/>

EDUCATION

2024–2028 **Harvard College**, Mechanical/Electrical Engineering S.B., cross-enrolled at MIT. GPA: 3.5/4.0
2020–2024 **Cambridge Rindge and Latin High School**, took classes at Harvard Extension School, Stanford Online, MITx. 2024 Massachusetts CTE Student of the Year. GPA: 3.8/4.0

Relevant Coursework: Advanced Intro to Robotics(Graduate Level); Circuits, Devices, and Transduction; Solid Mechanics; Mechanical Systems; Fluid Mechanics; Computer-Aided Machine Design; Intro to Computer Science; Multivariable Calculus; Linear Algebra & Differential Equations; Ordinary and Partial Differential Equations; Physics: Mechanics & Relativity, E&M, Waves, Electrodynamics

HIGHLIGHTED PERSONAL PROJECTS

Phase Array Sonar Laser

Nov 2025

- Designed custom circuitry and PCB, utilized STM32 processor and novel multiplexing latch setup to control precise phase, frequency, and amplitude of 64 independent signals. Overlayed audible tones on a 40KHz carrier frequency to transmit audible frequencies using HF piezo transducers and create a highly directional beam

Micro Swarming Drone

Jan 2026

- Creating an ultralight (52g) \$15 drone capable of automated landing/charging. Custom flight controller, drivers mounted on a <5cm board. Uses an ESP32 Processor and OV2640 onboard Camera.

E-Bike Conversion Kit

Dec 2023

- Budget-friendly design: custom PCB, Bluetooth control, BLDC BEMF detection using ATmega328P processor, custom motor driver circuit. Utilizes own 3D-printed hub motor design.

Homemade CNC Mill

Oct 2022

- Designed to rapidly prototype PCBs at home. Manufactured complex 2-sided PCBs for E-Bike, Home Control System, Go Kart, and other projects. 3D printed/8020 frame, stepper motors, Arduino for control

Go-Kart

Jul 2023

- Scrap steel frame, fly-by-wire, remote control, 212cc engine, self-designed auto-start, 3D-printed supercharger

RC Plane

Sep 2022

- Learned via successive failures of 9 separate designs and iterations. Foam airframe, FPV Camera, Servo Controlled. Mounted remotely activated homemade solid rocket motors to give extra boost.

Other Notable Projects Include: Gimbal Rocket Engine, Hybrid RC Car, 3D-Printed Gravity Clock, Smart Home Device, Arcade Cabinet, Hydroponics, RC Hot Wheels, Robot Vacuum, Stirling Engine, 400kV Ion Satellite Thruster
A full list of all of my projects can be found at: <https://grantkaufmann88.github.io/portfolio/>

EXPERIENCE

Engineering Intern, SpaceX

May 2026 – Aug 2026

- Worked on electrical integration team at Starlink Aviation, designed and qualified flight parts. Final Evaluation 4.1/5
- Had full ownership over qual of part used across fleet of 5,000+ aircraft, designed rig to simulate 45 years of life
- Had ownership of electrical integration on 500 Airbus A320s, personally surveyed aircraft in SAL, OKC, PI

Undergraduate Research Fellow, Rowland Institute

May 2024 – Present

- Sole research assistant under Winfield Hill, P.I., and Prof. Paul Horowitz, authors of *The Art of Electronics*
- Built sensitive geophones to record noise near electron microscopes, designed custom electromechanical actuators
- Developed PCB for a low-noise 500V vacuum tube power supply with custom hand-wound high-Q inductors
- Designed high-precision robotic angular positioning system for time-of-flight measurements at Vera Rubin Observatory

Vice President, Harvard Undergraduate Robotics Club

2024 – Present

- Lead a team of undergraduates to construct a Mars rover to compete in the University Rover Challenge
- Oversaw development of 6-DOF robotic arm, onboard testing lab, and AI navigation, assembled electronics. Taught new club members Circuit/PCB using a project-oriented approach.
- Grew membership from 10 to 60+ in one year. Applied Navy organizational management techniques and chain-of-command model to spread out responsibility across the small leadership team

Captain of Design, FIRST Robotics Competition

2023 – 2024

- Led 2024 robot design; managed team training and structure. Created and taught new course curriculum.
- Helped rebuild CRLS Engineering Department, served on hiring board. Taught Sophomore/Junior Engineering Classes.

SKILLS

Software

Onshape, Fusion 360, SolidWorks, Siemens NX, KiCAD, Eagle, Altium, PCB design, CAM, FEA, simulation, SPICE, COMSOL, C, C++, Python, Java, JavaScript, HTML/CSS, LaTeX, MATLAB

Manufacturing

Welding, soldering, carpentry, CNC, 3D printing, laser cutting, waterjet, lathe, OSHA 10 Certified

Other/Hobbies

Spanish (fluent, Seal of Billiteracy), long-distance hiking, canoeing, videography, rowing